Carbon Neutral Systems

REVIEW

Open Access

The ecology of artificial intelligence: energy, water, materials, and land limits of digital systems

Sibongiseni Hlabisa[1*]

Abstract

Artificial intelligence (AI) is promoted as a tool for ecological sustainability, yet the ecological effects of AI itself remain poorly understood. Existing studies document aspects of AI's footprint, including energy demand, water withdrawals, mineral extraction, and land occupation, but treat these impacts in isolation. This article develops an integrated resource nexus framework that situates AI within global environmental systems' energy–water–material–land metabolism. Drawing on recent evidence, we synthesize quantitative estimates of AI's resource intensity, from thousands of megawatt-hours and millions of liters required to train frontier models, to embodied mineral and land footprints embedded in semiconductor production and hyperscale data centers. We show that these demands are systemic, entangled, and spatially uneven, with benefits concentrated in the Global North and ecological costs externalized to resource frontiers in the Global South. The contribution of this article is twofold. First, it presents a novel conceptual model of AI as a socio-metabolic infrastructure, highlighting the entropic trade-offs associated with digital expansion. Second, it derives governance pathways, including AI-specific environmental impact assessments, multi-dimensional ecological metrics, and spatially responsible siting strategies that embed AI within planetary boundary frameworks. By reframing AI as an ecological actor with biophysical limits, we advance environmental systems research beyond narrow carbon accounting toward a holistic understanding of digital infrastructures in the Anthropocene.

Keywords Artificial Intelligence, Ecological effects, Energy–water–material–land Nexus, Digital infrastructures, Environmental governance, Planetary boundaries

1 Introduction

Artificial intelligence (AI) has become a defining feature of contemporary technological and environmental transitions. From climate modeling and biodiversity monitoring to energy optimization, AI is often portrayed as a cornerstone of sustainability innovation (Kaack et al. 2022; Lhoumeau, et al. 2025; Rolnick et al. 2022). However, beneath its digital abstraction lies a rapidly expanding physical infrastructure whose ecological footprint remains poorly understood.

Data centers, high-performance computing clusters, and semiconductor fabrication plants (fabs) consume substantial amounts of energy, water, materials, and land. These systems translate informational processing into material throughput, linking the digital economy to global patterns of extraction, pollution, and habitat loss. As Mytton (2021) and Ruberti (2024) show, the expansion of AI infrastructure has intensified local conflicts over electricity supply, water allocation, and land occupation, particularly in ecologically sensitive or water-scarce regions such as Arizona, Singapore, and Taiwan.

Despite the growing number of reports on AI's energy demand, few studies integrate its multi-resource metabolism, which encompasses the combined use of energy, water, materials, and land required to sustain its

*Correspondence:
Sibongiseni Hlabisa
hlabisb@unisa.ac.za
[1] Department of Environmental Sciences, University of South Africa, Roodepoort, Johannesburg, South Africa

Springer

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

computational operations. The absence of such integration hinders our ability to evaluate AI's actual ecological trade-offs and systemic vulnerabilities.

This article addresses this gap by situating AI within the resource nexus framework, which conceptualizes environmental pressures as interdependent rather than isolated. It adopts a socio-metabolic approach, combining quantitative synthesis and conceptual systems analysis to trace how digital infrastructures transform and appropriate biophysical resources. The analysis examines how much energy AI consumes, how water and land availability shape its spatial distribution, how material supply chains underpin its hardware production, and how these resource flows interact as an integrated system. It also examines the implications of these interconnections for environmental governance and the pursuit of sustainable digital transitions.

By integrating quantitative estimates with conceptual insights, this article provides one of the first comparative assessments of AI's ecological metabolism. Its broader aim is to clarify how the infrastructures enabling artificial intelligence both depend on and reshape the Earth's material foundations, and what this means for governing the digital age within planetary boundaries.

2 Methods

This study combines systematic literature synthesis with a socio-metabolic conceptual framework to examine the ecological dimensions of artificial intelligence. Rather than treating energy, water, materials, and land as separate categories of impact, the analysis approaches them as components of an interdependent system: a *resource nexus* through which AI's environmental footprint can be more accurately understood.

A systematic search was conducted across Web of Science, Scopus, and Google Scholar databases to identify studies published between 2015 and 2025 that quantified or analyzed AI-related environmental impacts. Search terms included combinations such as *"artificial intelligence" AND "energy consumption"*, *"data center water use"*, *"semiconductor fabrication environment"*, and *"data infrastructure land use"*.

After the initial search returned approximately 3,130,000 records, automated and manual screening were conducted in several stages. First, database filters were applied to restrict the search to English-language publications, peer-reviewed journals, and the period from 2015 to 2025, reducing the pool to approximately 18,300 studies. Second, keyword refinement was used to retain only studies explicitly addressing environmental aspects of artificial intelligence, such as *"energy use," "water footprint," "materials," "data centers,"* or *"semiconductor fabrication"*, further narrowing the sample to 312 records.

A title and abstract screening then excluded conceptual or social science papers that lacked empirical environmental data. The remaining 94 papers were assessed in full text, of which 56 met all inclusion criteria, namely: quantitative or model-based resource estimates, relevance to core AI infrastructures, and adequate methodological transparency for comparison.

After successive screening and eligibility checks, 46 sources were retained for analysis (see Fig. 1). The final corpus comprises 22 peer-reviewed publications (journal articles and refereed proceedings), 17 technical, policy, or industry reports (including government and corporate sustainability documents), four preprints/white papers, and three scholarly books or encyclopedia chapters. This mix balances empirical rigour and up-to-date technical insight, spanning foundational socio-metabolic and planetary boundaries research, quantitative studies of data centre energy and water use, and semiconductor manufacturing impacts. Together, these sources span the interdisciplinary landscape of environmental systems research, ranging from socio-metabolic theory (Fischer-Kowalski & Haberl 2007; Krausmann et al. 2008) and planetary boundaries (Rockström et al. 2009; Steffen et al. 2015) to contemporary analyses of data center energy and water consumption (LBNL 2024; Li et al. 2023; Mytton 2021) and semiconductor manufacturing impacts (Ruberti 2023, 2024).

The selected literature was analyzed through a qualitative–quantitative synthesis, extracting key metrics such as electricity demand per model training run, water withdrawal per megawatt-hour, embodied emissions per semiconductor unit, and average land footprint of hyperscale data centers and fabrication plants. Reported values were normalized to a common unit to facilitate cross-domain comparison.

Where possible, values were aggregated into representative ranges (e.g., 1.3–1.5 GWh per training run; 7,000–12,000 L per MWh for data center cooling) and cross-checked against recent environmental assessments (LBNL 2024; Mytton 2021; Ruberti 2023). These data were then contextualized through socio-metabolic reasoning, emphasizing how resource inputs and waste outputs circulate through AI's infrastructural systems.

2.1 Conceptual framework: the socio-metabolic nexus of AI

The analytical framework builds on the socio-metabolic theory of environmental systems (Fischer-Kowalski & Haberl 2007; Krausmann et al. 2008), which conceptualizes societies as energy–material systems sustained through continuous exchanges with the biosphere. This is integrated with the planetary boundaries framework (Rockström et al. 2009) to contextualize these flows

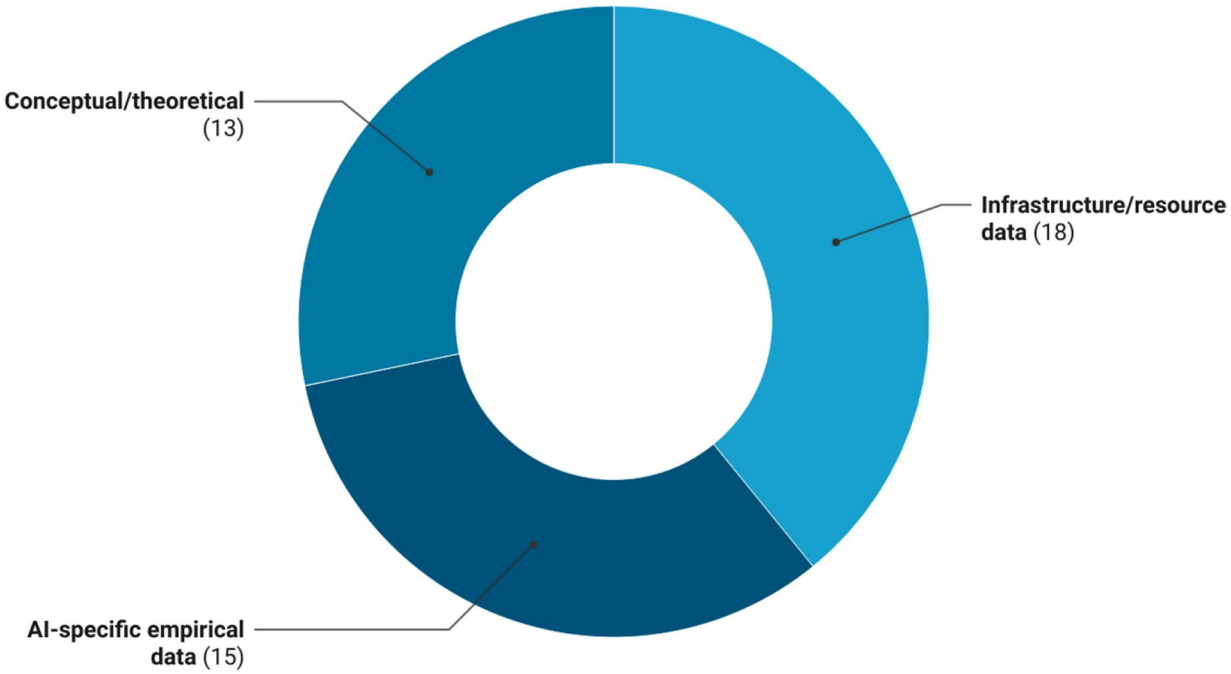


Fig. 1 Distribution of evidence types across 46 reviewed sources. Approximately one-third of studies provide empirical or model-based estimates specific to AI resource use, one-third report related infrastructural data, and the remainder offer conceptual or theoretical contributions

within Earth's safe operating space. Together, they offer a biophysical lens for analyzing AI's environmental relations. This framework highlights how computational infrastructures depend on the extraction, transformation, and spatial concentration of biophysical resources.

As illustrated in Fig. 2, AI functions as a *metabolic subsystem* that converts information into control capacity, sustained by flows of energy, water, and materials, and embedded within land-based infrastructure. The diagram captures both resource inputs (electricity, minerals, purified water) and entropic outputs (waste heat, pollution, and ecological displacement). This framing enables the analysis of AI as a technological and thermodynamically open subsystem of the global economy.

2.2 Limitations

While systematic in approach, this review is constrained by the scarcity and variability of available data. Many environmental estimates rely on modeled or proprietary values rather than empirical reporting, and few sources disaggregate AI-specific resource use from broader ICT infrastructure. Quantitative synthesis therefore, emphasizes representative orders of magnitude rather than precise totals. Despite these limitations, the integration of multiple data types and sources enables a coherent view of AI's emergent ecological profile and highlights critical areas for future empirical research.

3 Results: the resource nexus of AI

The synthesis of 46 selected sources reveals that artificial intelligence infrastructures exhibit substantial and interlinked demands for energy, water, materials, and land. These dependencies are mutually reinforcing, forming a socio-ecological metabolism in which each resource dimension intensifies the others. The findings are organized below around these four domains and their cross-domain interactions.

3.1 Overview of resource intensities

Across all studies, AI's ecological metabolism emerges as a resource-dense and spatially concentrated phenomenon. As shown in Table 1, training a single large model, such as GPT-3 or PaLM-2, can consume between 1.3 and 1.5 gigawatt-hours (GWh) of electricity, roughly equivalent to the annual energy usage of a small town (Patterson et al. 2021). At the global scale, total AI-related electricity demand could reach 150–300 TWh yr^{-1} by 2028, a three- to five-fold increase from 2022 (LBNL 2024).

Water withdrawals show a comparable magnitude. Data-center cooling typically requires 7,000–12,000 L MWh^{-1}, while semiconductor fabrication uses 1.8–2.5 m^3

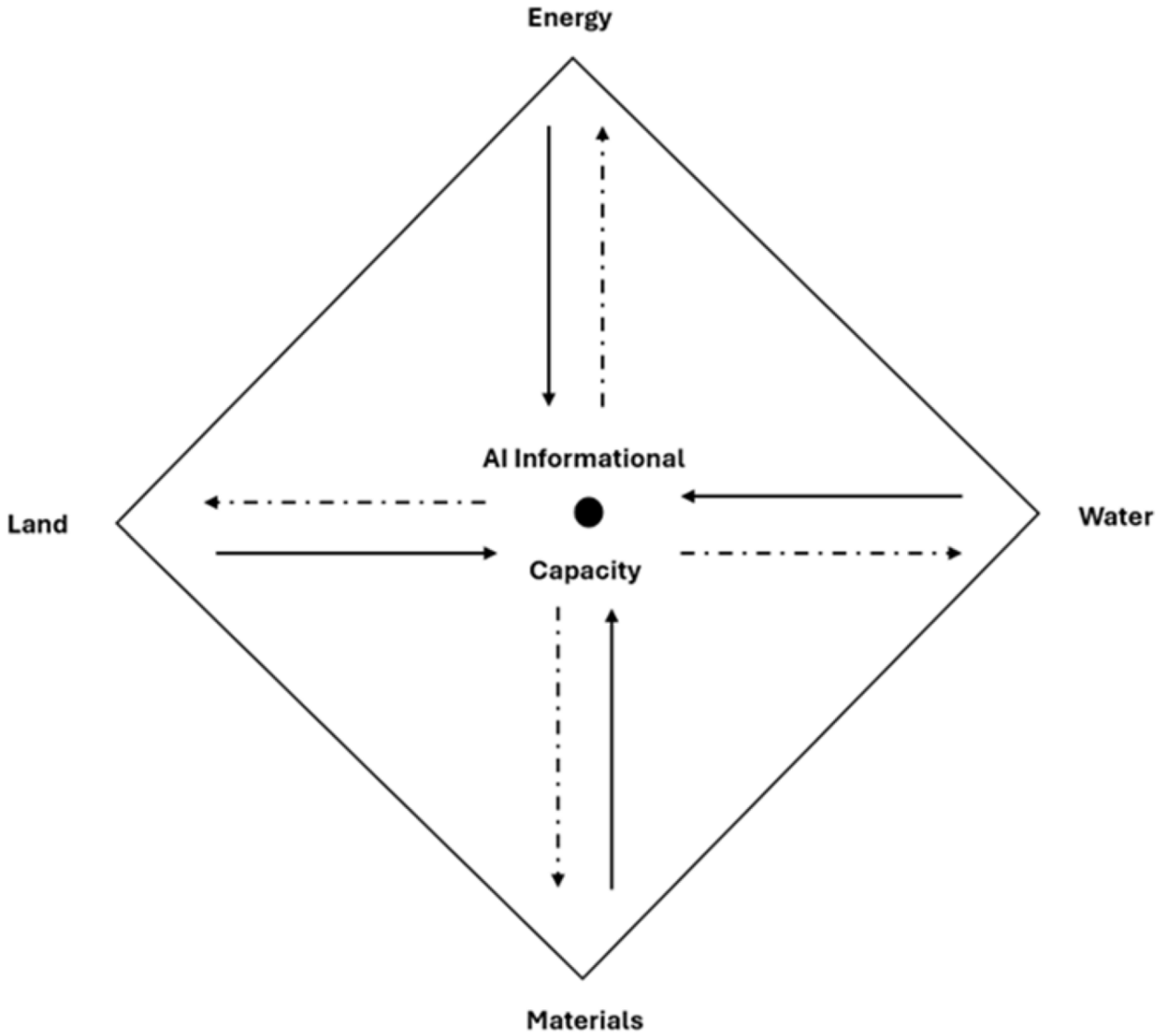


Fig. 2 The ecological footprint of AI as a resource nexus. Solid arrows indicate inputs and dashed arrows indicate entropic outputs

H_2O per wafer, often as ultrapure water (Mytton 2021; Ruberti 2023). Each NVIDIA HGX H100 baseboard emits approximately 1.3 t CO_2-equivalent, with more than 90% of emissions arising from the production of materials and components (NVIDIA 2022b). Spatially, a hyperscale data center occupies 30–100 ha, and a modern semiconductor fab complex extends over 20–60 ha (Roundy 2025; Ruberti 2024).

These data demonstrate how AI translates information processing into physical throughput across multiple resource systems.

3.2 Energy

Energy constitutes the central driver of AI's environmental metabolism. The rapid scaling of model size and computational workload has significantly increased electricity consumption in data centers worldwide. Although successive generations of processors have achieved efficiency gains, Google (2024) reports a more than 30-fold improvement in energy per inference between 2023 and 2024. However, aggregate demand continues to rise, confirming the persistence of rebound effects (Sorrell 2009). These trends suggest that improvements in efficiency

Table 1 Quantitative estimates of AI's resource footprint

Domain	Indicator	Estimated Range/Typical Value	Illustrative Sources
Energy	Electricity use per large-model training (e.g., GPT-3, PaLM-2)	1.3–1.5 GWh per model run	International Energy Agency 2025; Patterson et al. 2021; Strubell et al. 2019
	Global AI-related electricity demand (projected 2028)	150–300 TWh yr^{-1}	LBNL 2024; Shehabi et al. 2024
	Energy intensity improvement (2023–2024)	> 30 × reduction per inference	Google 2024
Water	Data-center cooling water use	7,000–12,000 L MWh^{-1}	Mytton 2021
	Semiconductor fabrication water use	1.8–2.5 m^3 H_2O per wafer produced	Ruberti 2023; TSMC 2022
	AI-training operational water footprint	2–5 L per 1,000 inferences (approx.)	Li et al. 2023
Materials	Embodied carbon per HGX H100 baseboard	≈ 1.3 t CO_2-eq (cradle-to-gate)	NVIDIA 2022b; Schneider et al. 2025
	Key materials required	High-purity Si, Cu, Co, Ta, Nd, Tb	Ali et al. 2017; Wübbeke 2013
	Semiconductor by-products	$SiCl_4$, HF, and GHG emissions	Ruberti 2024
Land/Space	Land footprint of hyperscale data center	30–100 ha per site	Roundy 2025; Monstadt & Saltzman 2025
	Land footprint of semiconductor fab complex	20–60 ha per facility	Ruberti 2024; TSMC 2022
	Locations facing land-use constraints	Ireland and Singapore (moratoria and conditional permits)	Government of Ireland 2022; DataCenter Dynamics 2022

alone cannot decouple AI development from increasing energy consumption, demonstrating the need for systemic rather than technological interventions.

The projected trajectories in Fig. 3 reveal both the substantial scale and significant uncertainty of future electricity demand from artificial intelligence. According to the International Energy Agency's (2025) analysis, global consumption is projected to grow steadily through 2030 across all scenarios. However, the magnitude varies considerably depending on assumptions about AI

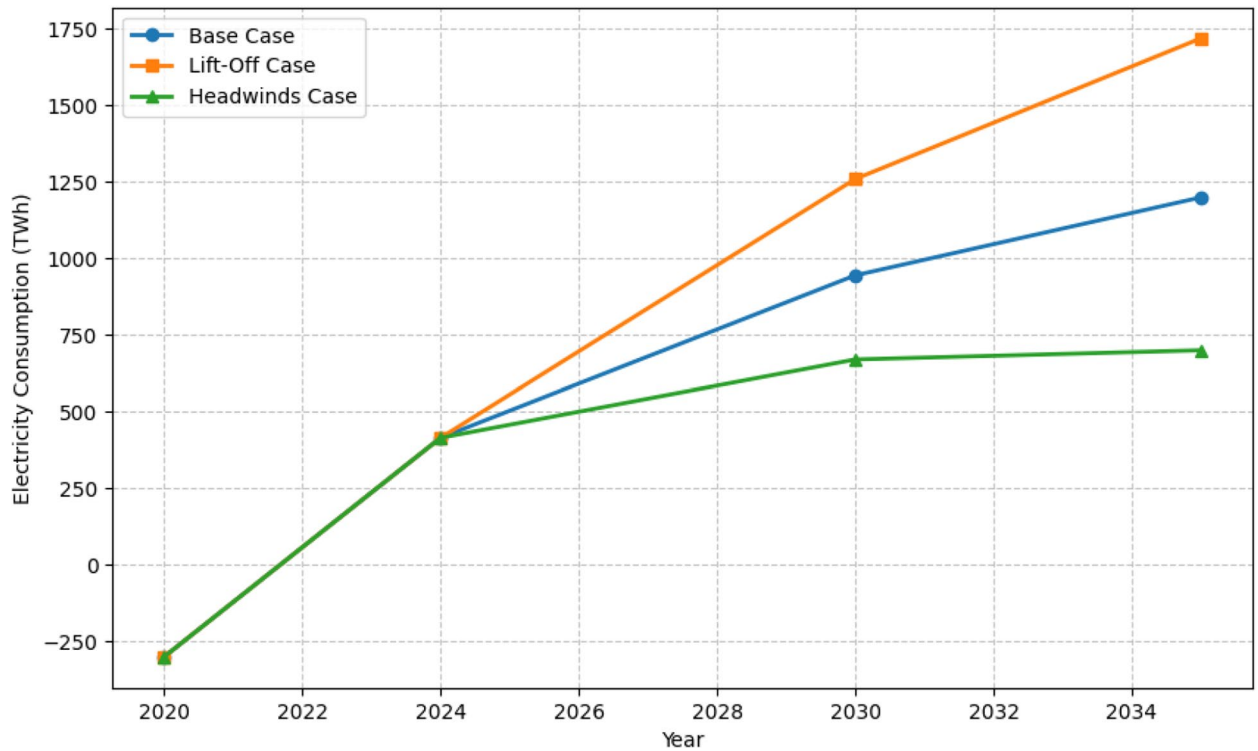


Fig. 3 Global electricity demand from data centers shows significant uncertainty across scenarios. In the Base Case, demand more than doubles from 415 TWh in 2024 to 945 TWh by 2030, reaching ~ 1,200 TWh by 2035. The Lift-Off Case (stronger AI uptake, fewer constraints) reaches over 1,700 TWh by 2035, while the High Efficiency and Headwinds cases show lower trajectories of ~ 970 TWh and ~ 700 TWh, respectively

adoption rates, computational intensity, and improvements in energy efficiency. The Base Case projects continued growth that would more than double data center electricity consumption by 2030. In contrast, the Lift-off scenario, characterized by accelerated AI deployment and fewer constraints, shows demand reaching levels comparable to major industrial sectors. Conversely, the Headwinds scenario illustrates how regulatory barriers, economic challenges, or supply chain limitations could moderate this growth, albeit resulting in substantially higher consumption than in 2022 (International Energy Agency 2025).

The geographical distribution of AI-related electricity demand demonstrates significant concentration effects. As illustrated in Fig. 4, the United States accounts for nearly half of the global growth in data center electricity consumption, followed by China. In contrast, Europe exhibits more constrained expansion due to its regulatory frameworks (IEA 2025). This spatial clustering creates distinct energy demand hotspots, presenting complex challenges for grid management and environmental sustainability.

3.3 Water

Water plays a dual role in AI infrastructure, serving as both a cooling agent and an enabler of precision manufacturing. In data centers, water usage is influenced by the local climate and cooling design, with evaporative systems typically prevailing in arid regions. In semiconductor fabrication, the production of ultrapure water multiplies the embodied energy of each wafer, creating a hidden hydrological burden. In areas such as Arizona, Singapore, and Taiwan, competition among industrial, municipal, and agricultural users has already led to episodes of conflict and vulnerability to drought (Mytton 2021; Narvaez 2022). From a socio-metabolic perspective, water functions as a thermodynamic sink, allowing the dissipation of computational heat and linking digital performance directly to ecological stress.

The scenario analysis (Fig. 5) reveals that AI's future water footprint is not predetermined but hinges critically on the interplay between adoption rates and efficiency gains. The "Rapid AI Adoption" scenario illustrates a concerning pathway where a 15-fold increase in compute load, accompanied by minimal efficiency improvements (5%), could lead to a more than 1,000% increase in water

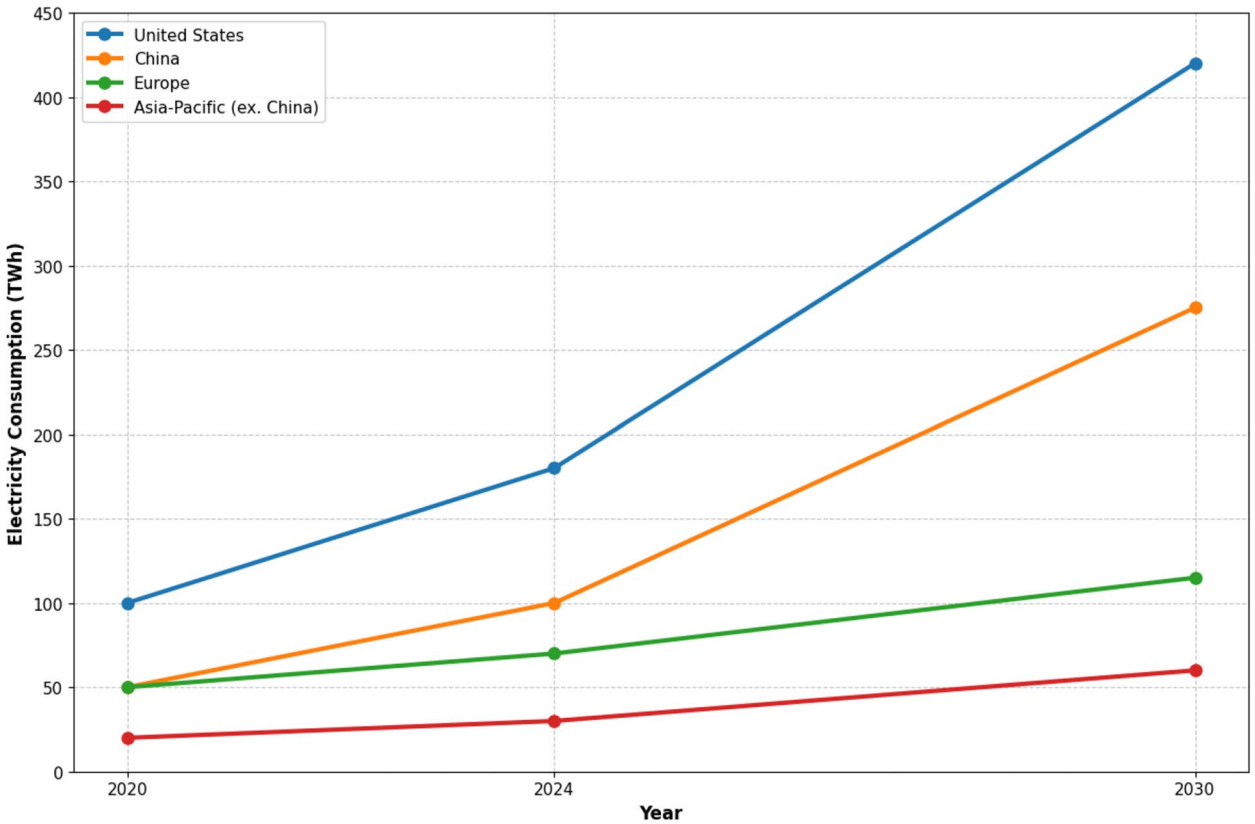


Fig. 4 The United States and China dominate the growth in data center electricity consumption. In the Base Case, the US accounts for nearly half of the global increase, with consumption rising by ~ 240 TWh from 2024 to 2030. China follows as the second-largest market, adding ~ 175 TWh over the same period. Europe and other regions show significant but comparatively lower growth

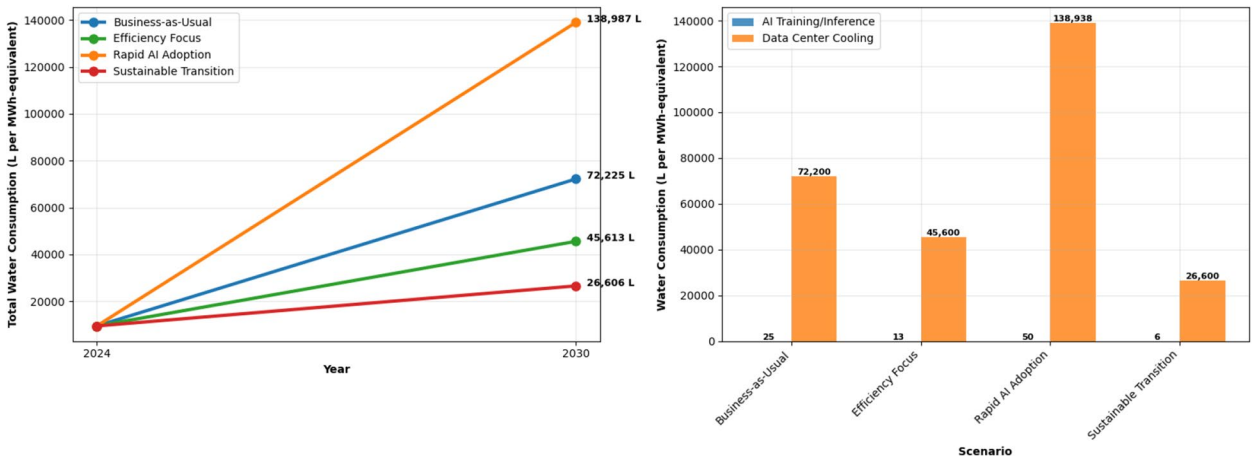


Fig. 5 Scenario analysis of AI-related water consumption (2024–2030). Left: Growth trajectories under different adoption and efficiency assumptions. Right: 2030 consumption breakdown by component. The "Rapid AI Adoption" scenario projects a more than tenfold increase in water usage. In contrast, the "Sustainable Transition" scenario shows that aggressive efficiency measures (resulting in a 60% improvement) could limit growth to approximately four times current levels, highlighting the critical role of water-efficient AI development

consumption by 2030. This trajectory would position AI infrastructure as a significant industrial water consumer, potentially straining resources in water-scarce regions where data centers are concentrated. Conversely, the "Sustainable Transition" scenario illustrates that aggressive investment in water-efficient cooling technologies and optimized AI models could decouple AI growth from resource consumption, limiting the increase to approximately 300% despite a fourfold expansion in AI workloads. The pronounced variation between scenarios highlights the urgent need for policy interventions and technological innovations aimed at enhancing the water efficiency of AI systems and data centers.

3.4 Materials

Material use anchors AI's dependence on extractive industries. Advanced processors and memory chips require refined silicon, copper, cobalt, tantalum, and rare-earth elements such as neodymium and terbium. Their extraction generates extensive ecological externalities, including toxic waste and riverine contamination (Ali et al. 2017; Wübbeke 2013). Within fabrication, hazardous by-products such as silicon tetrachloride pose additional pollution risks if not recycled or neutralized (Thomas 2019).

Ruberti (2023, 2024) observes that despite improvements in eco-efficiency, the environmental intensity of chip production has plateaued because each generation demands cleaner rooms and greater throughput. This pattern illustrates an entropic escalation: maintaining a higher informational order requires a disproportionate amount of material and energetic input, deepening AI's entanglement with global supply chains and contributing to environmental degradation.

Figures 6 and 7 provide a visual illustration of the "entropic escalation" inherent in AI's physical infrastructure.

Figure 6 quantifies the central paradox outlined by Ruberti (2023, 2024): despite incremental improvements in eco-efficiency, the resource intensity of chip manufacturing surges with each new generation. The left panel clearly shows this divergence, as the industry progresses from 28 to 2 nm nodes, eco-efficiency plateaus, while the demand for resources skyrockets. This occurs because the gains from technical efficiency are overwhelmed by the escalating costs of purity and precision, such as the need for vast clean rooms and ultra-pure materials. The right panel further contextualizes this issue, revealing that AI-optimized chips are not marginally, but substantially, more resource-hungry, demanding up to 4.5 times more water and rare earth elements than traditional chips. This chart captures the essence of entropic escalation: achieving a higher state of informational order (more powerful AI) requires a disproportionately larger input of material and energy.

Figure 7 expands this linear progression into a systemic model, illustrating how the environmental impacts of AI hardware form a self-reinforcing cycle. The circular flow demonstrates that the consequences are not confined to a single stage but are interconnected. For instance, the hazardous e-waste resulting from disposal (a consequence of the complex chips shown in Fig. 6) can lead to soil and water contamination, which in turn exacerbates the ecological damage and resource scarcity associated with the initial extraction phase. This creates a feedback loop

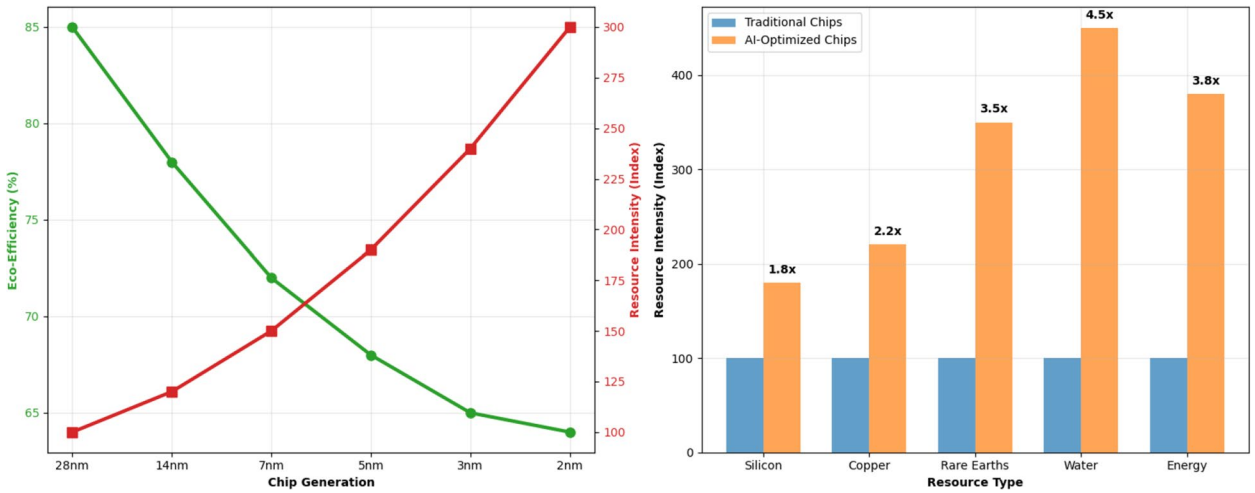


Fig. 6 Efficiency plateau and rising resource intensity across chip generations. AI-optimized hardware shows 1.8–4.5 × higher material demands than traditional chips, demonstrating entropic escalation

where solving one environmental problem is hampered by the consequences of another. The diagram thus moves beyond a simple supply chain to show why the environmental challenge is so persistent and difficult to solve; the system itself is structured to perpetuate and intensify its own resource demands and ecological footprint.

3.5 Land and space

AI's land footprint reflects the physical manifestation of its infrastructure, a transformation vividly captured in the accompanying figures. Figure 8 provides a stark visual testimony to this phenomenon, mapping how clusters of data centers and fabs transform peri-urban and agricultural landscapes into specialized industrial zones. The land-use charts quantify this shift, showing how regions like Ireland and Singapore have seen AI infrastructure claim a significant portion of their land, directly displacing agricultural and other uses. This global snapshot contextualizes the specific policy responses you cite; the map clearly shows Ireland and Singapore categorized under 'Moratorium' and 'Conditional' status, a direct result of them hitting the capacity limits acknowledged in their government statements.

The long-term implications of this spatial transformation are explored in Fig. 9. It powerfully illustrates the concept of path dependency, demonstrating how large semiconductor clusters, such as Hsinchu Science Park and Arizona's corridor, lock land into industrial function for decades. The infrastructure life-cycle chart illustrates a multi-decade "lock-in" period, resulting in the very spatial inflexibility that constrains alternative ecological or social uses. Furthermore, the trade-off analysis explains *why* this inflexibility is so critical: while AI infrastructure scores high on economic capacity, it scores lowest on ecological value and flexibility, the very qualities needed for sustainable and adaptive land management.

Finally, Fig. 10 offers a systemic diagnosis of the limits being reached. Its multi-dimensional analysis shows that the moratorium in Ireland and the conditional lift in Singapore were not arbitrary but were triggered by a convergence of critical constraints. These regions, along with other hotspots such as Virginia, are shown pushing against red-line thresholds not just in land allocation, but also in grid capacity, water stress, and regulatory pressure simultaneously. This figure moves beyond anecdotal evidence to provide an integrated model that explains why the reconfiguration of regional land use creates such complex and binding constraints on future development.

3.6 Nexus interactions

The four resource domains interact in non-linear ways. Higher energy use increases cooling demand, which in turn leads to increased water withdrawals. Water scarcity can restrict energy production and chip fabrication, both of which depend on material extraction that transforms landscapes. These feedbacks form a coupled resource-nexus system, in which local efficiency gains may produce global environmental costs.

Understanding AI's ecological footprint, therefore, requires moving beyond single-metric accounting toward an integrated, systemic analysis. The nexus perspective highlights that the combined pressures of energy availability, water scarcity, material throughput, and land use constrain the growth of AI. These interactions set the stage for the discussion that follows, which interprets the

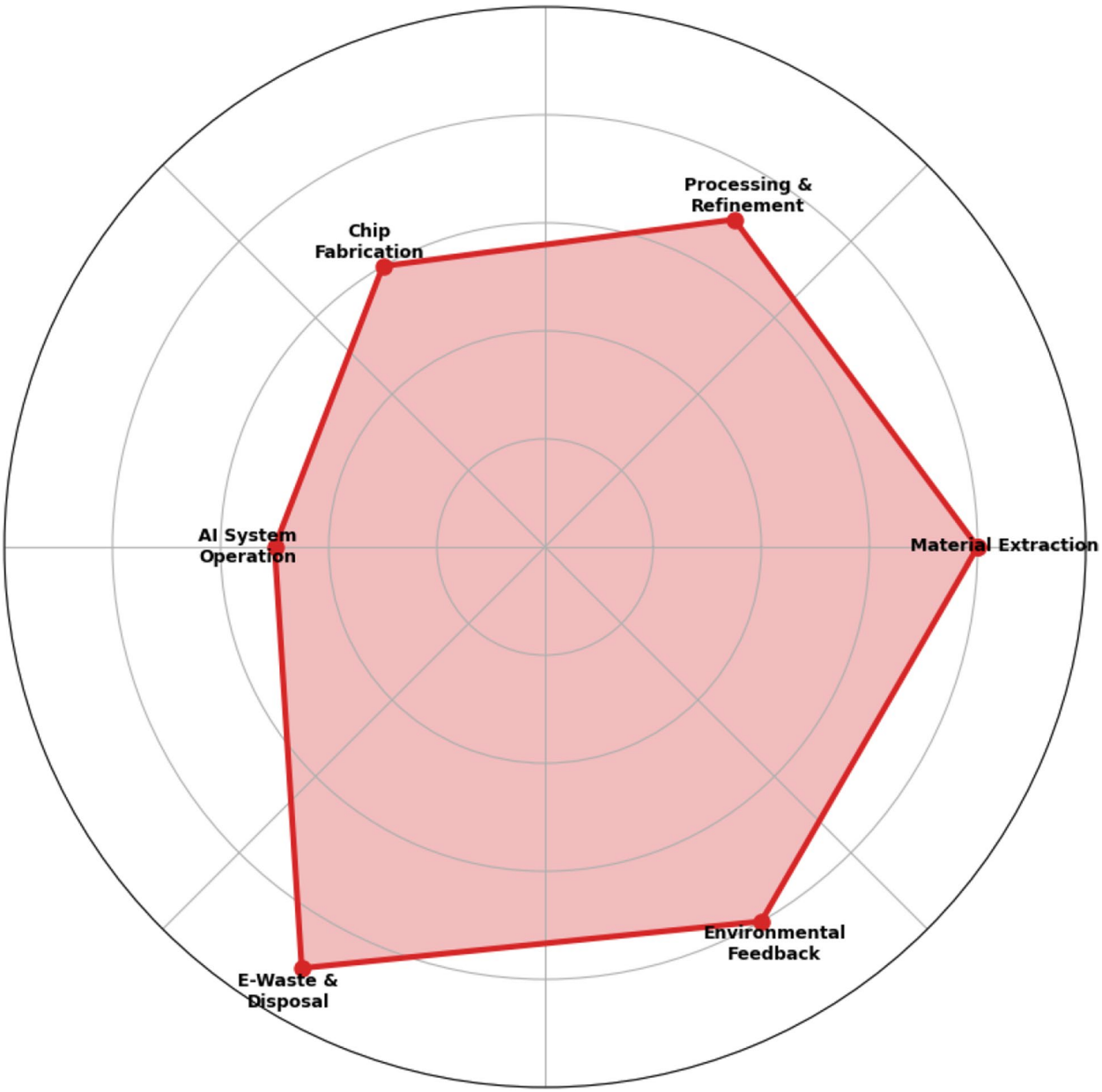


Fig. 7 Circular environmental impact model showing interconnected lifecycle stages and feedback loops that drive entropic escalation in AI hardware systems. Each stage feeds back into environmental degradation, creating a self-reinforcing cycle of resource intensity

implications for biodiversity, resilience, and environmental governance.

4 Discussion

The synthesis of forty-six sources reveals that the environmental footprint of artificial intelligence (AI) extends well beyond electricity consumption. AI operates through a multi-resource metabolism in which energy, water, materials, and land are tightly coupled. This interdependence produces both *efficiency gains* and *systemic vulnerabilities*. Despite rapid technical improvements, the broader pattern suggests a deepening resource dependence that scales faster than gains in efficiency or substitution can offset.

4.1 Complexity, efficiency, and the paradox of control

A recurring pattern across the reviewed literature is the contradiction between technological efficiency and aggregate environmental load, a dynamic often captured by the Jevons paradox (Sorrell 2009). As AI architectures

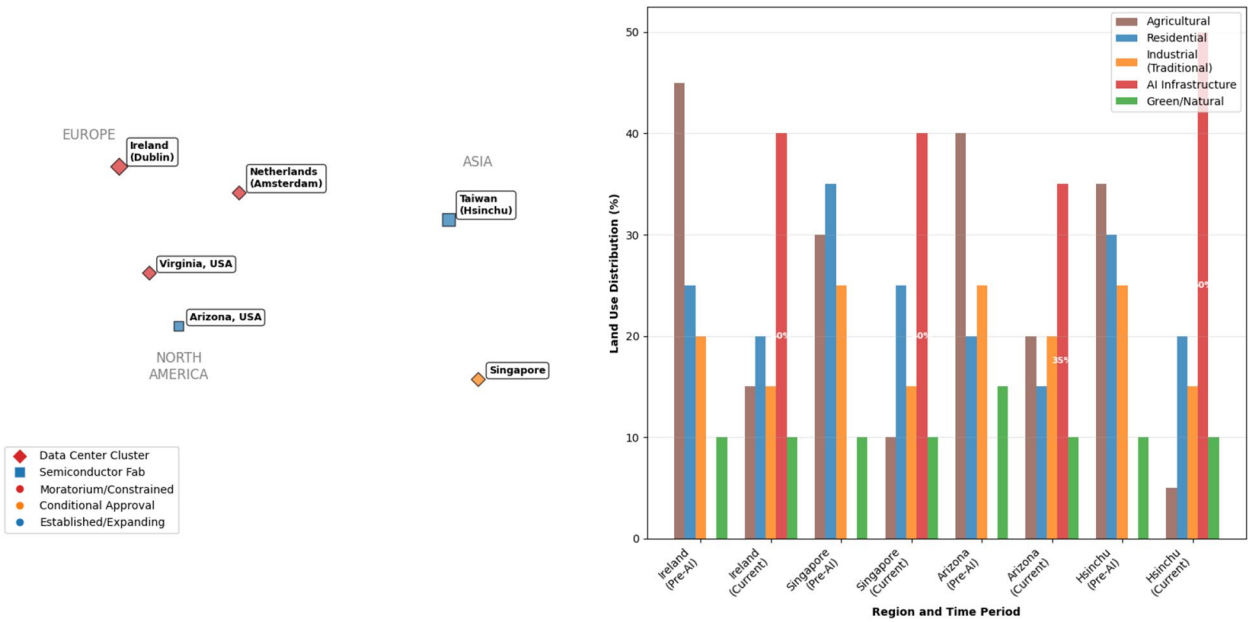


Fig. 8 Global AI infrastructure hotspots and their transformative impact on regional land use, showing significant conversion of agricultural and residential land to industrial AI clusters

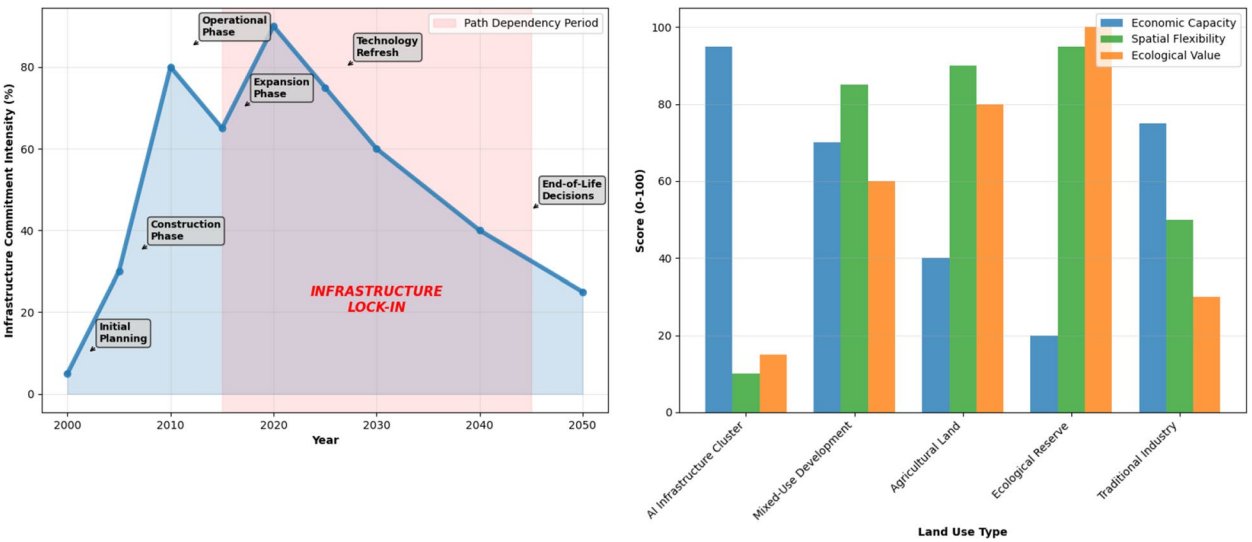


Fig. 9 Infrastructure lifecycle showing decades-long "lock-in" period and comparative analysis of land use trade-offs, revealing AI infrastructure's high economic value but low ecological and flexibility scores

evolve, each generation of hardware and software becomes markedly more efficient at processing information. Google's 2024 *Environmental Sustainability Report*, for example, documents more than a 30-fold reduction in energy required per inference within just a single year of optimization. Similar trends are visible in NVIDIA's shift from the A100 to the H100 architecture, where performance per watt has increased dramatically (NVIDIA 2022a; 2022b). Yet, as these improvements lower the marginal cost of computation, they stimulate broader deployment of AI services, expanding both total computational throughput and the physical infrastructure that supports it.

This feedback produces what might be called an efficiency–complexity spiral. Each incremental improvement in efficiency enables the design of larger models,

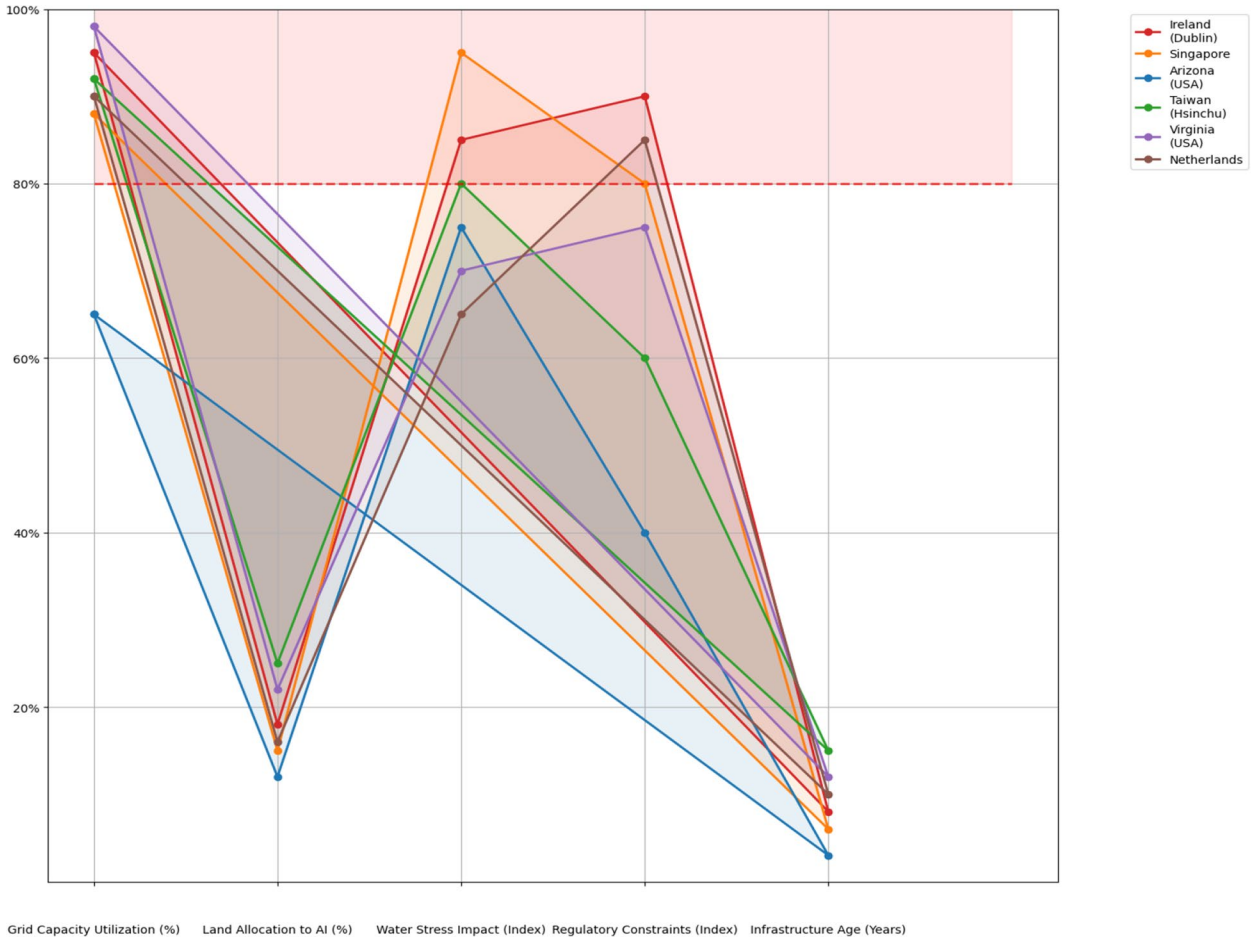


Fig. 10 Multi-constraint saturation analysis of AI infrastructure hubs, revealing critical pressure points in grid capacity, land allocation, and regulatory frameworks across regions

denser data centres, and more pervasive applications. In doing so, efficiency gains are quickly reinvested into greater system complexity. Empirical evidence from the data-center sector confirms this pattern: although the average energy use per computation has fallen, global electricity consumption by data centers continues to grow at roughly 15–20 percent annually (LBNL 2024). Water consumption follows a similar trajectory, as cooling demand scales with total thermal output, even as individual servers become more efficient (Mytton 2021). The apparent decoupling of performance from environmental cost thus proves temporary; aggregate demand outpaces efficiency.

From a thermodynamic perspective, this outcome reflects a fundamental constraint of open systems. Maintaining and expanding informational order requires a continuous inflow of energy and matter, along with an outflow of waste heat and entropy. The more intricate the informational structure, be it a neural network or a planetary economy, the greater the energetic throughput required to sustain it. In this sense, the pursuit of control through computation inevitably deepens material dependence. The growth of AI, therefore, parallels the broader pattern of industrial civilization described in socio-metabolic theory (Fischer-Kowalski & Haberl 2007; Krausmann et al. 2008): progress in informational complexity drives expansion in biophysical metabolism.

The paradox of control lies in the fact that systems seeking greater control over their environment must expend increasing amounts of energy to maintain that very control (Hlabisa 2025). In AI, this is manifested in escalating requirements for data storage, redundancy, cybersecurity, and algorithmic refinement, all of which consume additional computational cycles and hence additional energy. The outcome is a self-reinforcing loop: the more AI is used to optimize energy systems, logistics, or agriculture, the more computational infrastructure is required to support these optimizations. What emerges is not a linear path toward dematerialization,

but a circular process in which efficiency generates new domains of dependency.

This insight challenges prevailing narratives of "green AI," which often assume that algorithmic optimization will automatically yield sustainability benefits. While local efficiencies are real and sometimes substantial, they are embedded within a global system whose total energetic and material demands are rising. The policy implication is that sustainability cannot be achieved solely through efficiency improvements; it requires explicit constraints on total computational growth and a shift from efficiency-based metrics to sufficiency-based governance. Without such systemic limits, the energetic metabolism of AI will continue to accelerate even as individual devices become cleaner or faster.

4.2 Spatial concentration and systems vulnerability

The global infrastructure sustaining artificial intelligence is characterized by extraordinary spatial concentration. Production, computation, and storage are concentrated within a limited number of industrial and climatic nodes, notably Taiwan, the United States, Ireland, and Singapore, each of which hosts large-scale semiconductor fabrication plants or hyperscale data centres. This pattern mirrors broader trends in socio-metabolic systems, where technological specialization yields efficiency at the cost of geographical resilience. As Monstadt and Saltzman (2025) observe, the "digital gateways" of Europe are physically materialized in a handful of infrastructural corridors, which concentrate energy flows, land conversion, and risk exposure.

The Taiwan Semiconductor Manufacturing Company (TSMC) offers a striking example. Its operations in Hsinchu Science Park alone account for billions of liters of annual freshwater use, extensive electricity demand, and significant waste heat emissions (TSMC 2022; Ruberti 2024). During the 2021 drought, water shortages compelled authorities to reallocate supplies from agriculture to semiconductor production (Narvaez 2022), demonstrating how AI's physical backbone directly competes with food systems and rural livelihoods. A similar dynamic is emerging in Arizona's semiconductor corridor, where water withdrawals for new fabs are provoking public concern (Lepawsky 2024). These cases highlight how digital infrastructures are not only dependent on natural resources but also actively reshape their allocation.

In Europe, and particularly in Ireland, the concentration of data centers has led to tensions between industrial growth and public sustainability goals. The Government of Ireland's (2022) statement on data centers explicitly acknowledges that energy grid constraints and water availability limit further expansion, resulting in a temporary moratorium on new developments. Singapore has adopted comparable measures, imposing a construction freeze in 2019 and subsequently allowing only conditional approvals tied to efficiency and heat-recovery performance (DataCenter Dynamics 2022). Together, these examples illustrate how spatial saturation and resource competition are emerging as critical governance frontiers for digital economies.

Spatial concentration also amplifies systemic vulnerability. In resilience theory, systems that centralize critical functions in a few nodes exhibit high efficiency but low redundancy (Pisano 2012). In the AI sector, concentration magnifies exposure to environmental hazards, supply chain disruptions, and geopolitical shocks. A drought in Taiwan, a heatwave in Ireland, or an energy crisis in Europe can have a ripple effect across global information flows, potentially disrupting AI model training, cloud services, and dependent sectors such as finance, logistics, or environmental monitoring. From a socio-metabolic perspective, this pattern resembles entropy localization, where the accumulation of environmental and thermodynamic stress occurs in a few sites, allowing informational order to be maintained elsewhere.

The geographic asymmetry of this system reflects the externalization logic of global production. High-value informational outputs are generated in the global North, while many of the ecological and material costs are absorbed by resource-intensive industrial nodes in Asia and the Middle East (Ali et al. 2017; UNEP 2020). This asymmetry reproduces older patterns of environmental inequality under a new digital guise. While AI applications are often promoted as tools for global optimization (enhancing crop yields, monitoring biodiversity, or improving logistics), their infrastructural underpinnings remain materially and spatially exclusionary, requiring secure energy and water access that many regions cannot afford.

This concentration further undermines systemic resilience. The principle of redundancy, which involves maintaining multiple, geographically distributed systems to buffer against local shocks, is largely absent in AI infrastructure design. Instead, hyperscale data centers and fabrication clusters seek economies of scale through aggregation, producing single points of environmental and geopolitical failure. As LBNL (2024) and Shehabi et al. (2024) note, the majority of global data processing is now handled by fewer than one hundred sites, each consuming gigawatt-scale power. The vulnerability of such a system lies in its environmental exposure and its dependence on continuous flows of energy, materials, and cooling resources, which climate change and political instability increasingly threaten.

A broader implication is that digital concentration is becoming a new form of ecological risk. Just as global agriculture has become vulnerable through monocultures, the digital economy's concentration in a few industrial ecologies introduces fragility on a planetary scale. This suggests a need to rethink spatial governance for AI infrastructure, including decentralizing data processing, diversifying geographic locations, and integrating local renewable energy and water resources. Policies that mandate environmental impact assessments, resilience planning, and equitable resource allocation can help prevent AI's material concentration from translating into systemic collapse.

From the standpoint of socio-metabolic theory, the pattern is unmistakable: AI's expansion reproduces the metabolic geography of industrial modernity, in which the control of information depends on the extraction and concentration of energy and matter in limited spaces. The apparent intangibility of artificial intelligence thus rests on intensely territorial and ecological foundations.

4.3 Biodiversity, land systems, and hidden ecological costs

While the environmental discourse on artificial intelligence (AI) has largely focused on energy consumption and carbon emissions, the land-use and biodiversity dimensions remain insufficiently recognized. However, as the reviewed literature shows, AI's infrastructure exerts growing pressure on terrestrial ecosystems through land occupation, habitat fragmentation, and indirect resource-driven degradation. In effect, the expansion of digital infrastructures (data centers, semiconductor foundries, transmission corridors, and supporting logistics) represents a new phase of industrial territorialization, translating informational growth into physical reconfiguration of landscapes (Rosati et al. 2023; Monstadt & Saltzman 2025).

Data centers and semiconductor fabrication facilities are spatially extensive, often requiring tens to hundreds of hectares of serviced land equipped with cooling systems, substations, and access infrastructure (Roundy 2025; Ruberti 2024). Once established, these facilities are long-lived and spatially inert, remaining operational for decades. Their establishment typically entails the permanent conversion of agricultural, peri-urban, or semi-natural landscapes. In Ireland, for instance, community opposition to hyperscale data centers has increased in response to land-use conflicts and cumulative energy demand (Government of Ireland 2022). Similarly, Singapore's early moratorium on data center expansion reflected concerns about limited land availability and ecosystem stress in its already densely populated urban environment (DataCenter Dynamics 2022).

This form of land use exemplifies what could be described as metabolic enclosure, the appropriation and stabilization of land for high-throughput industrial systems. Once enclosed, such land is effectively withdrawn from alternative ecological or social functions. Over time, this produces spatial lock-in, whereby environmental governance becomes constrained by the sunk costs of infrastructural investments. Land that might otherwise serve as agricultural buffer zones, wetland restoration sites, or community commons becomes fixed within a high-energy digital economy.

AI's land footprint is not limited to the built environment; it extends upstream through resource extraction and supply chains. Semiconductor manufacturing depends on high-purity silicon, copper, cobalt, and rare earth elements such as neodymium and terbium (Ali et al. 2017; Wübbeke 2013). The mining and processing of these materials often occur in ecologically sensitive areas, such as China, Africa, and South America, resulting in deforestation, soil contamination, and riverine pollution. For instance, tailings and acid mine drainage from rare-earth processing in Inner Mongolia have created toxic waste lakes, which have destroyed local vegetation and compromised groundwater quality (Wübbeke 2013).

Such upstream degradation represents a hidden land cost; a displacement of ecological burden from high-income, infrastructure-consuming regions to resource-producing peripheries. This externalization of impact reinforces global environmental inequality, whereby the biophysical consequences of AI's growth are disproportionately borne by ecosystems and communities far removed from centers of consumption.

Habitat conversion associated with industrial and urban expansion remains one of the primary drivers of biodiversity decline globally (IPBES 2019). The siting of AI infrastructure often coincides with peri-urban biodiversity hotspots, due to its proximity to transport corridors and grid infrastructure. Large industrial estates alter microclimates, disrupt ecological connectivity, and contribute to urban heat islands, further stressing local flora and fauna.

Moreover, the intensification of resource extraction for AI hardware compounds this loss. The mining and smelting of critical minerals involve the destruction of habitats, the introduction of invasive species through road networks, and long-term toxic effects on aquatic ecosystems (Edwards et al. 2014). In biodiversity terms, the AI industry therefore acts as both a direct agent of land transformation and an indirect amplifier of extractive frontiers, accelerating the rate at which natural systems are converted into industrial metabolism.

The dematerialized image of digital technology often obscures these spatial and ecological impacts. The

perception of AI as a low-footprint, knowledge-based innovation masks its physical dependence on continuous flows of land, water, and materials. The invisibility of these processes within environmental accounting systems, particularly in nations that host high-end computational infrastructure, perpetuates a policy blind spot. Land-use planning frameworks rarely incorporate digital infrastructure into biodiversity impact assessments, and environmental impact statements typically focus on construction-phase emissions rather than operational or cumulative ecosystem effects.

From a socio-metabolic perspective, this invisibility is symptomatic of an information–material asymmetry: the more energy and matter are consumed to sustain digital order, the more abstracted and immaterial that order appears. This paradox of immateriality constitutes a profound challenge for environmental governance. Without explicit recognition of AI's terrestrial footprint, policies risk reinforcing the very disconnection between knowledge production and ecological reality that drives unsustainable development.

Recognizing the land dimension of AI's metabolism opens up new pathways for sustainability policy. Incorporating digital infrastructures into strategic environmental assessments, establishing spatial caps on land conversion for data and fabrication sites, and developing ecological offset mechanisms could help internalize these costs. Likewise, adopting biodiversity-sensitive design principles, such as vegetated cooling systems, landscape buffers, and integration with renewable energy corridors, can mitigate local impacts.

However, such measures must be guided by a systemic understanding: land conservation cannot be achieved by simply relocating environmental burdens elsewhere. As Seto et al. (2012) emphasize, the sustainability of urban and industrial systems depends on recognizing land teleconnections: the distant ecological linkages that connect production, consumption, and environmental change across scales. In this light, AI's expansion is not merely a technological phenomenon but a spatial and ecological process that demands an integrative governance response.

4.4 Water–energy–material nexus: from efficiency to sufficiency

A central insight from this review is that the environmental burden of artificial intelligence (AI) cannot be understood through any single variable (energy, water, or materials) taken in isolation. Instead, these three domains interact as a resource nexus that defines the thermodynamic and ecological metabolism of AI infrastructures. Each input reinforces or constrains the others: electricity generation depends on water for cooling; semiconductor fabrication consumes both energy and ultrapure water; and the production of advanced materials requires extensive energy and chemical processing. The overall effect is a *multiplicative coupling* of environmental costs, where local improvements in one dimension may worsen pressures in another.

Data centers illustrate this nexus most clearly. As computational loads increase, more servers and cooling equipment are required, resulting in higher electricity and water consumption. Facilities that switch from air to evaporative cooling systems to improve energy efficiency often increase their water usage (Mytton 2021). Conversely, water-free cooling solutions such as closed-loop chillers can increase energy demand by 10–15 percent (Shehabi et al. 2024). These trade-offs are context-specific: in arid regions such as Arizona or the Gulf states, electricity generation itself relies on thermoelectric plants that withdraw vast volumes of water, compounding local scarcity (Yang & Lant 2011). Thus, reducing one environmental metric often amplifies another, creating governance dilemmas that are more about substitution than resolution.

The semiconductor sector magnifies this coupling. A single 300-mm wafer may require up to 2.5 m^3 of ultrapure water and several hundred kilowatt-hours of electricity (Ruberti 2023; TSMC 2022). Purification and recirculation systems depend on chemical reagents and high-pressure pumps, which themselves consume energy. At the same time, the chemicals used in wafer cleaning (hydrofluoric acid, hydrogen peroxide, and silicon tetrachloride) entail significant embodied emissions and toxicity risks (Shen et al. 2018). The production, transportation, and disposal of these materials form a continuous loop of energy-material exchange that sustains the informational order of the digital economy.

Within this coupled nexus, efficiency gains often fail to reduce aggregate throughput. Improvements in chip architecture or data-center design lower the resource cost per computation but stimulate larger computational demand. As seen with the transition from NVIDIA's A100 to H100 accelerators, each watt saved per operation enables the proliferation of more servers and training runs (NVIDIA 2022b; Schneider et al. 2025). The same logic applies to water: reductions in cooling water intensity invite higher server density and greater total heat generation. This rebound effect, which illustrates why efficiency-enabling growth remains inadequate when sustainability strategies focus solely on technological optimization (Sorrell 2009), is a key example.

A more robust approach requires shifting the governing principle from efficiency to sufficiency, the deliberate limitation of total resource use within ecological boundaries. In practical terms, this means that efficiency metrics

(e.g., power usage effectiveness or water usage effectiveness) must be complemented by absolute consumption caps. Data centers, for example, could be licensed based on both carbon-intensity targets and total water-use quotas tied to regional hydrological capacity. Semiconductor fabs may adopt integrated resource accounting, reporting cumulative energy and water intensity per output unit, including the embedded costs of upstream materials.

Sufficiency also implies temporal coordination: aligning AI workloads with periods of renewable-energy surplus or cooler climatic conditions to minimize peak stress. Emerging concepts such as "follow-the-sun" or "follow-the-wind" computing (Edinger 2024; Vergallo et al. 2025), where data tasks migrate geographically in response to renewable availability, illustrate how computational demand can adapt to ecological rhythms rather than override them.

A shift toward sufficiency is inextricably linked to the principles of a circular economy. Recovering heat from data-center operations for district energy, recycling ultrapure water within fabrication plants, and reclaiming critical minerals from discarded hardware all contribute to lowering the system's net entropy. However, such measures remain peripheral within the AI industry, where competitive pressures prioritize performance over circularity. Policies that mandate resource-recovery benchmarks and material traceability could transform these isolated innovations into systemic norms.

In this sense, sufficiency is not austerity; it is intelligent restraint: a design philosophy that recognizes thermodynamic limits as opportunities for creativity and resilience. By reframing progress as the maintenance of informational order with minimal ecological disruption, the AI sector could evolve from a linear growth paradigm toward a regenerative one.

Ultimately, governing the water–energy–material nexus requires institutional coordination that transcends sectoral silos. Environmental regulation remains fragmented: water policy seldom considers data-center cooling; energy planning rarely accounts for material extraction; and industrial standards for semiconductors overlook downstream e-waste. Integrating these domains under a nexus-based governance framework would allow policymakers to evaluate cumulative impacts and design cross-resource trade-off mechanisms, for instance, permitting increased energy use only when offset by verified material or water savings.

Such frameworks could be aligned with planetary boundary thinking (Rockström et al. 2009; Steffen et al. 2015), establishing quantitative safe operating limits for AI's metabolic flows. Implementing them would not merely reduce environmental pressure; it would transform AI from a passive consumer of ecosystem services into an active participant in sustaining them.

The cross-sectoral nature of artificial intelligence (AI) presents a formidable governance dilemma. Unlike traditional industries, AI's environmental impacts are dispersed across multiple domains: energy, water, materials, and land, each regulated under different institutions, jurisdictions, and data regimes. This fragmentation mirrors the very structure of AI's infrastructure: distributed globally, privately owned, and largely opaque. The result is a systemic accountability gap between the scale of environmental consequences and the scope of existing governance mechanisms.

Currently, there is no international framework that systematically tracks the biophysical costs of digital technologies. Data on energy and water use are primarily self-reported by corporations, and material footprints are typically limited to confidential supply-chain audits. Public datasets, such as those provided by the IEA (2021) or UNEP (2020), aggregate at the level of national economies rather than individual infrastructures. This opacity prevents policymakers and researchers from accurately evaluating AI's actual ecological burden.

Industry sustainability reports (e.g., Google 2024; NVIDIA 2022b) often highlight efficiency improvements but omit absolute consumption figures, masking overall growth in throughput. This selective disclosure exemplifies what Hornborg (2001) described as the *illusion of technological autonomy*: the notion that advanced systems can be evaluated independently of their material dependencies. The absence of standardized environmental accounting protocols for AI not only hinders regulation but also perpetuates public misperception of digital technologies as inherently "clean" or "dematerialized."

AI's physical infrastructure cuts across conventional governance boundaries. Semiconductor manufacturing in Taiwan, data storage in Ireland, algorithm development in the United States, and user demand in Africa or Latin America all form a single transnational system. However, environmental regulation remains largely territorial and sectoral, addressing discrete activities within national borders. This jurisdictional mismatch renders environmental oversight both fragmented and reactive.

Even when national environmental impact assessments are required, they often exclude cumulative and cross-resource impacts. For instance, construction approvals for data centers in Dublin or Amsterdam may assess carbon emissions but neglect embedded water withdrawals or land displacement. Similarly, export controls on critical minerals do not extend to end-use accountability for AI hardware. As a result, governance operates like a patchwork quilt stitched across a global industrial

metabolism, with no central mechanism to manage its total entropy.

Adding to the complexity is the corporate concentration of AI capacity. A handful of multinational corporations (Google, Microsoft, Amazon, NVIDIA, and TSMC) dominate the hardware, cloud, and chip-manufacturing sectors that underpin global AI development. This concentration gives private actors disproportionate influence over environmental outcomes and the transparency of data. It also allows corporations to define the parameters of sustainability on their own terms.

As Kaack et al. (2022) note, most corporate sustainability commitments focus on operational emissions ("Scope 2") while excluding embodied impacts and external supply chains ("Scope 3"). This narrow focus perpetuates a form of *technocratic environmentalism* that prioritizes efficiency within corporate boundaries rather than systemic sufficiency across the entire AI ecosystem. Without mandatory disclosure and external verification, voluntary corporate governance remains structurally insufficient to align AI development with ecological limits.

Efforts to introduce environmental metrics for AI, such as model energy scores, water usage effectiveness (WUE), and power usage effectiveness (PUE), represent progress toward accountability but risk metric reductionism. By quantifying only what is measurable, such indicators can obscure unmeasured externalities, including biodiversity loss, material toxicity, and spatial displacement. Moreover, efficiency metrics can easily be co-opted into green marketing rather than systemic reform. As Sorrell (2009) warns, governance through efficiency metrics may reproduce rebound effects if not coupled with absolute limits on consumption.

Nevertheless, standardized metrics remain essential as a first step. Integrating them into Environmental, Social, and Governance (ESG) reporting, science-based target frameworks, or ISO environmental standards could improve comparability across firms. Extending these standards to cover data centers, chip manufacturing, and cloud service providers would establish a robust empirical foundation for international regulation.

The complexity of AI's socio-metabolic footprint necessitates what Stirling (2014) terms reflexive governance, an approach that acknowledges uncertainty, incorporates multiple knowledge systems, and emphasizes adaptive learning over command-and-control regulation. Reflexive governance recognizes that environmental oversight must evolve in tandem with technological innovation. In the context of AI, this implies building institutions capable of cross-sector coordination, participatory decision-making, and dynamic boundary setting.

One promising direction is the creation of a Global Observatory for Digital Resource Use, modeled on existing bodies such as the IPCC or IPBES, but focused on the ecological dimensions of the digital economy. Such an institution could compile standardized data on the energy, water, and material flows of AI systems, assess their alignment with planetary boundaries, and support evidence-based policymaking. At the national level, integrating AI resource metrics into environmental accounting systems, for example, the UN's System of Environmental-Economic Accounting (SEEA), could link digital expansion to physical sustainability indicators.

Another approach is polycentric governance, where overlapping institutions, international organizations, national agencies, and local authorities share responsibility for monitoring and managing AI's resource metabolism. This distributed model mirrors the very architecture of AI itself, allowing for adaptive oversight without central control. By linking corporate transparency requirements, local environmental licensing, and international reporting standards, polycentric governance could transform fragmented oversight into a coordinated framework for ecological accountability.

Ultimately, the governance of AI's environmental impacts must address more profound ethical and epistemological questions. Who decides what counts as "sustainable AI"? Whose data, energy, and water sustain the digital systems used to predict, optimize, or govern others? As Floridi and Cowls (2019) argue, ethical AI requires fair algorithms and just infrastructures, systems whose material foundations reflect distributive equity and ecological care. Embedding justice into environmental governance thus means extending accountability beyond emissions to encompass the entire socio-metabolic chain.

From a systemic perspective, accountability is both a regulatory function and a thermodynamic necessity. Information and control, as Wiener (1948) observed, require energy; therefore, every act of environmental oversight entails its own metabolic cost. The challenge of governance in the age of AI is to minimize the entropy of control itself, to design institutions that maintain ecological order with the least possible expenditure of energy, data, and political complexity. Achieving this balance is the ultimate test of sustainable intelligence.

4.5 Research gaps and future directions

Despite rapid growth in the literature on artificial intelligence (AI) and sustainability, this review reveals that empirical knowledge of AI's resource metabolism remains sparse and uneven. Of the forty-six sources synthesized, only about one-third provide quantitative data directly measuring or modelling AI-related energy, water, material, or land use. The remainder extrapolates from related infrastructure or provides conceptual frameworks

without empirical substantiation. This imbalance highlights a critical need for systematic data generation and methodological integration across the environmental and computational sciences.

The first and most urgent gap concerns empirical quantification. Few peer-reviewed studies disaggregate AI resource use from that of general information and communication technology (ICT). Energy-use data are often proprietary, while water and material intensities of AI hardware remain opaque. Semiconductor and cloud-computing companies release sustainability reports (e.g., Google 2024; NVIDIA 2022b; TSMC 2022), yet these reports provide relative metrics rather than absolute consumption figures. Independent verification is rare.

Future research must therefore develop transparent measurement protocols for AI infrastructure, analogous to the monitoring systems used in industrial ecology or energy accounting. This could include (1) *life-cycle assessments (LCAs)* of model training and inference phases, (2) *real-time metering* of data-center resource flows, and (3) *harmonized units* for cross-comparing energy, water, and material intensities. Establishing shared reporting standards would not only support academic research but also underpin regulatory frameworks and corporate accountability.

A second research frontier involves methodological integration. Environmental systems analysis and computer science still operate in parallel rather than interactive domains. While environmental scholars quantify physical throughput, computer scientists optimize algorithmic performance. Bridging these perspectives demands the development of hybrid models that explicitly couple computational complexity with resource flows, linking bits to joules, and operations to emissions.

One promising direction is the incorporation of socio-metabolic and thermodynamic principles into digital-systems modelling. By mapping the energetic and material cost per computational operation, researchers can construct a *digital material flow analysis* (d-MFA) that situates AI within global metabolism. Such work could draw on the tradition of Fischer-Kowalski and Haberl (2007) and Krausmann et al. (2008), expanding socio-metabolic theory into the informational domain.

Similarly, advances in complex-systems simulation offer opportunities to model feedbacks between resource efficiency, rebound effects, and governance interventions. Integrating agent-based models with empirical data could illuminate the thresholds at which technological complexity turns from adaptive to destabilizing, a central concern for sustainability science.

Current research rarely disaggregates environmental impacts by geography or social context. However, as demonstrated above, the burdens of AI infrastructure, namely energy demand, water withdrawals, and land conversion, are concentrated in a few regions, whereas the benefits are more diffuse globally. This unequal spatial distribution raises questions of environmental justice and global governance that remain underexplored in the AI literature.

Future studies should map resource supply chains from extraction to disposal, identifying the communities and ecosystems most affected. Incorporating satellite data, corporate disclosures, and trade statistics could reveal how the informational economy redistributes environmental pressure across space and class. This would extend the emerging field of *digital political ecology*, grounding debates on AI ethics in material and geographic realities.

The environmental analysis of AI also requires deeper theoretical grounding. Present debates oscillate between techno-optimism and techno-fatalism, with little understanding of the physical laws that govern informational order. Thermodynamics provides a unifying framework: all systems that process information must expend energy and externalize entropy. Building a formal theory of information thermodynamics in governance and technology could link disparate disciplines, cybernetics, systems ecology, and industrial metabolism into a coherent basis for environmental systems research.

Such a theory would move beyond descriptive metrics toward predictive principles: how much energy is required to sustain a given degree of informational control? When does optimization become counterproductive? Addressing these questions could help design AI architectures and governance mechanisms that maintain function without exceeding ecological thresholds.

Finally, progress will depend on institutional coordination. Academic research, corporate disclosure, and policy design remain siloed. Establishing cross-sector consortia that bring together environmental scientists, engineers, economists, and policymakers would accelerate knowledge integration. Initiatives such as a Global Observatory for Digital Resource Use or dedicated funding streams for "AI and the Environment" within major science agencies could institutionalize this field.

Equally important is building human capacity: training environmental scientists in computational methods and computer engineers in sustainability assessment. Such interdisciplinary literacy would enable the next generation of researchers to quantify and govern AI's ecological metabolism holistically.

The future of AI sustainability research hinges on a shift from fragmented case studies to systemic, multi-scalar, and interdisciplinary inquiry. Quantitative evidence must be expanded; theoretical foundations must be deepened; and governance mechanisms must evolve

in parallel. Doing so will transform AI from an opaque consumer of resources into a domain of experiment for sustainable design where intelligence, in both human and artificial forms, becomes an instrument for ecological balance rather than its erosion.

5 Conclusion

This review has shown that artificial intelligence (AI), far from being an abstract or "weightless" technology, is deeply embedded in the biophysical fabric of the Earth system. Across forty-six interdisciplinary sources, the evidence demonstrates that AI infrastructures, comprising data centers, semiconductor fabrication plants, and global supply chains, constitute a new layer of industrial metabolism characterized by high energy throughput, intensive water use, material extraction, and expanding land use. These four domains are not discrete; they are mutually reinforcing components of a single socio-ecological process through which information is produced, stabilized, and circulated.

The results reveal a persistent efficiency–complexity paradox: while each generation of hardware and algorithms becomes more efficient, aggregate resource demand continues to rise. Technological improvement thus coexists with expanding environmental pressure, mirroring broader socio-metabolic transitions observed throughout industrial history. The expansion of AI therefore, exemplifies a new phase of the global resource economy where the optimization of information depends on the intensification of material flows.

Spatially, AI's metabolism is concentrated and uneven. Industrial clusters in Taiwan, the United States, Singapore, and Ireland have become strategic nodes in the digital economy, but they are also sites of water stress, energy strain, and land-use conflict. The environmental footprint of AI is thus not universal; it is territorially and politically situated, reflecting global asymmetries in production and consumption. Such concentration enhances efficiency but erodes resilience, exposing the world's informational systems to ecological and geopolitical shocks.

The review also highlights the hidden ecological costs of AI, particularly biodiversity loss and upstream degradation associated with the mining and refining of rare materials. The illusion of digital dematerialization conceals a vast network of extractive frontiers, waste streams, and spatial enclosures that transform natural landscapes into industrial metabolism. Recognizing these hidden dimensions is essential for re-embedding digital innovation within the boundaries of ecological regeneration.

From a systems perspective, AI's environmental challenges arise not merely from technological inefficiency but from the structure of governance itself. Current regulatory regimes are fragmented, jurisdictionally mismatched, and dependent on voluntary corporate disclosure. Effective environmental governance will require integrated, reflexive, and polycentric frameworks that treat AI as part of a global resource nexus rather than a discrete sector. Mechanisms such as standardized reporting, global observatories, and nexus-based licensing could align digital growth with ecological limits.

For research, the findings point toward an emerging field of digital socio-metabolic analysis: a convergence of environmental systems science, thermodynamics, and information theory. Future work should advance empirical quantification, spatial mapping, and theoretical modelling to reveal how computation interacts with material and energetic constraints. This will demand collaboration across disciplines, from environmental scientists and engineers to social theorists and policymakers.

Ultimately, the question is not whether AI can be made more efficient, but whether intelligence – human or artificial – can evolve toward sufficiency and ecological balance. True sustainability will depend on re-imagining intelligence itself as a form of stewardship: the ability to maintain informational order without destabilizing the natural systems that sustain it. Recognizing AI as a material, metabolic, and moral phenomenon is therefore not a limitation of technology but its next frontier.

Acknowledgements
Not applicable.

Authors' contributions
S.H. is the sole author of this work. The author conceived the study, developed the theoretical framework, conducted the analysis, prepared all figures, and wrote the manuscript.

Funding
The author received no financial support for this research.

Data availability
The study is based on publicly available secondary data and literature sources cited throughout the manuscript. No new datasets were generated or analyzed during the current study. All materials and data supporting the findings are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 20 October 2025 Revised: 13 November 2025 Accepted: 17 November 2025
Published online: 02 December 2025

References

Ali SH, Giurco D, Arndt N, Nickless E, Brown G, Demetriades A et al (2017) Mineral supply for sustainable development requires resource governance. Nature 543:367–372. https://doi.org/10.1038/nature21359

DataCenter Dynamics. (2022). Singapore lifts data center moratorium but sets conditions. DataCenter Dynamics. Available at: https://www.datacenterdynamics.com/en/news/singapore-lifts-data-center-moratorium-but-sets-conditions/

Edinger, J. (2024). Follow-The-Sun Computing – Merging Green Energy with Smart Computing. ERCIM News, 138. Available at: https://www.inf.uni-hamburg.de/en/inst/ab/dos/newslist/2023-10-05-follow-the-sun-first-big-step.html.

Edwards DP, Sloan S, Weng L, Dirks P, Sayer J, Laurance WF (2014) Mining and the African environment. Conserv Lett 7(3):302–311. https://doi.org/10.1111/conl.12076

Fischer-Kowalski M, Haberl H (2007) Socio-ecological transitions and global change: Trajectories of social metabolism and land use. Edward Elgar, Cheltenham

Floridi, L., & Cowls, J. (2019). A unified framework of five principles for AI in society. Harvard Data Sci Rev, 1(1). https://doi.org/10.1162/99608f92.8cd550d1

Google. (2024). Environmental sustainability report. Mountain View: Google. Available at: https://sustainability.google/reports/google-2024-environmental-report/

Government of Ireland. (2022). Government statement on the role of data centres in Ireland's enterprise strategy. Department of Enterprise, Trade and Employment. Available at: https://www.gov.ie/en/department-of-enterprise-tourism-and-employment/publications/government-statement-on-the-role-of-data-centres-in-irelands-enterprise-strategy/

Hlabisa S (2025) The control-entropy paradox: modeling the thermodynamic limits of environmental governance. Syst Res Behav Sci. https://doi.org/10.1002/sres.3178

Hornborg A (2001) The power of the machine: global inequalities of economy, technology, and environment. AltaMira Press, Walnut Creek

International Energy Agency (IEA) (2025) Energy and AI. IEA, Paris

International Energy Agency (IEA). (2021). The role of critical minerals in clean energy transitions. Paris: IEA. Available at: https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions

IPBES. (2019). Global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. Brondizio, E. S., Settele, J., Díaz, S., & Ngo, H. T. (eds). Bonn: IPBES Secretariat. https://doi.org/10.5281/zenodo.3831673

Kaack LH, Quiroz K, Callaway D et al (2022) Aligning artificial intelligence with climate change mitigation. Nat Clim Chang 12:518–527. https://doi.org/10.1038/s41558-022-01377-7

Krausmann F, Fischer-Kowalski M, Schandl H, Eisenmenger N (2008) The global sociometabolic transition: past and present metabolic profiles and their future trajectories. J Ind Ecol 12(5–6):637–656. https://doi.org/10.1111/j.1530-9290.2008.00065.x

Lawrence Berkeley National Laboratory (LBNL). (2024). U.S. data center energy outlook. Berkeley: LBNL. https://doi.org/10.71468/P1WC7Q

Lepawsky J (2024) Climate change induced water stress and future semiconductor supply chain risk. iScience 27(2):108791. https://doi.org/10.1016/j.isci.2024.108791

Lhoumeau S, Pinelo J, Borges PA (2025) Artificial intelligence for biodiversity: exploring the potential of recurrent neural networks in forecasting arthropod dynamics based on time series. Ecol Indic 171:113119. https://doi.org/10.1016/j.ecolind.2025.113119

Li, P., Yang, J., Islam, M. A., & Ren, S. (2023). Making AI less "thirsty": Uncovering and addressing the secret water footprint of AI models. arXiv preprint. https://arxiv.org/abs/2304.03271

Monstadt J, Saltzman K (2025) How data centers have come to matter: Governing the spatial and environmental footprint of the 'Digital Gateway to Europe.' Int J Urban Reg Res 49(4):757–778. https://doi.org/10.1111/1468-2427.13316

Mytton D (2021) Data centre water consumption. NPJ Clean Water 4:869–871. https://doi.org/10.1038/s41545-021-00101-w

Narvaez, L. (2022). Technical Report: Taiwan drought – Interconnected disaster risks 2021/2022. UNU Collections. Available at: https://collections.unu.edu/view/UNU:9027

NVIDIA. (2022a). NVIDIA H100 tensor core GPU architecture: Exceptional performance, scalability, and security for the data center. Technical White Paper. NVIDIA Corp. Available at: https://resources.nvidia.com/en-us-hopper-architecture/nvidia-h100-tensor-c

NVIDIA. (2022b). NVIDIA HGX H100 baseboard: Product carbon footprint summary. NVIDIA Corporation. Available at: https://images.nvidia.com/aem-dam/Solutions/documents/HGX-H100-PCF-Summary.pdf

Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L. M., Rothchild, D., … Dean, J. (2021). Carbon emissions and large neural networks. arXiv preprint arXiv:2104.10350. https://doi.org/10.48550/arXiv.2104.10350

Pisano, U. (2012). Resilience and Sustainable Development. ESDN Quarterly Report No. 26. European Sustainable Development Network. Available at: https://www.esdn.eu/fileadmin/ESDN_Reports/2012-September-Resilience_and_Sustainable_Development.pdf

Rockström J, Steffen W, Noone K, Persson Å, Chapin FS, Lambin E et al (2009) Planetary boundaries: Exploring the safe operating space for humanity. Nature 461:472–475. https://doi.org/10.1038/461472a

Rolnick D, Donti PL, Kaack LH, Kochanski K, Lacoste A, Sankaran K et al (2022) Tackling climate change with machine learning. ACM Comput Surveys 55(2):1–96. https://doi.org/10.1145/3485128

Rosati C, James A, Metcalf K (2023) Data plantation: Northern Virginia and the territorialization of digital civilization in 'the Internet Capital of the World.' Online Media and Global Communication 2(2):199–227. https://doi.org/10.1515/omgc-2023-0017

Roundy, J. (2025). The increasing concern of data center land acquisition. TechTarget. Available at: https://www.techtarget.com/searchdatacenter/feature/The-increasing-concern-of-data-center-land-acquisition

Ruberti M (2023) The chip manufacturing industry: environmental impacts and eco-efficiency analysis. Sci Total Environ 858(Part 2):159873. https://doi.org/10.1016/j.scitotenv.2022.159873

Ruberti M (2024) Environmental performance and trends of the world's semiconductor foundry industry. J Ind Ecol 28(5):1183–1197. https://doi.org/10.1111/jiec.13529

Schneider, I., Xu, H., Benecke, S., Patterson, D., Huang, K., Ranganathan, P., & Elsworth, C. (2025). Life-cycle emissions of AI hardware: A cradle-to-grave approach and generational trends. arXiv preprint. https://arxiv.org/abs/2502.01671

Seto KC, Reenberg A, Boone CG, Fragkias M, Haase D, Langanke T et al (2012) Urban land teleconnections and sustainability. Proceed Natl Acad Sci 109(20):7687–7692. https://doi.org/10.1073/pnas.1117622109

Shehabi, A., Smith, S. J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M. A. B., Holecek, B., Koomey, J., Masanet, E., & Sartor, D. (2024). United States data center energy usage report. Berkeley: Lawrence Berkeley National Laboratory. https://doi.org/10.71468/P1WC7Q

Shen C, Tran P, Pham L (2018) Chemical waste management in the U.S. semiconductor industry. Sustainability 10(5):1545. https://doi.org/10.3390/su10051545

Sorrell S (2009) Jevons' paradox revisited: the evidence for backfire from improved energy efficiency. Energy Policy 37(4):1456–1469. https://doi.org/10.1016/j.enpol.2008.12.003

Steffen, W., Richardson, K., Rockström, J., Cornell, S. E., Fetzer, I., Bennett, E. M., et al. (2015). Planetary boundaries: Guiding human development on a changing planet. Science, 347(6223). https://doi.org/10.1126/science.1259855

Stirling, A. (2014) From sustainability to transformation: dynamics and diversity in reflexive governance of vulnerability. SWPS 2014–06. https://doi.org/10.2139/ssrn.2742113

Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (pp. 3645–3650). Stroudsburg, PA: Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1355

Thomas, L. (2019). The Environmental Impacts of Photovoltaic Technology. AZoCleanTech. Available at: https://www.azocleantech.com/article.aspx?ArticleID=831

TSMC. (2022). 2022 sustainability report. Taiwan Semiconductor Manufacturing Company, Taipei. Available at: https://www.tsmc.com/english/about TSMC/dc_csr_report

United Nations Environment Programme (UNEP). (2020). Mineral resource governance in the 21st century: Gearing extractive industries towards

sustainable development. Nairobi: UNEP. Available at: https://www.unep.org/resources/report/mineral-resource-governance-21st-century
Vergallo, R., Cruz, L., Errico, A., & Mainetti, L. (2025). On the Effectiveness of the 'Follow-the-Sun' Strategy in Mitigating the Carbon Footprint of AI in Cloud Instances [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2506.10990
Wiener, N. (1948). Cybernetics: Or Control and Communication in the Animal and the Machine. MIT Press.
Wübbeke J (2013) Rare earth elements in China: policies and narratives of supply chains. Resour Policy 38(3):384–394. https://doi.org/10.1016/j.resourpol.2013.05.005
Yang X, Lant C (2011) A geographical and temporal analysis of U.S. thermoelectric withdrawals in comparison to renewable water supplies. Appl Geogr 31(3):1154–1165. https://doi.org/10.1016/j.apgeog.2011.03.006

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.